

Diagnosing Hardware Failures Theory

Task 1: 5 Common Desktop PC Hardware Issues and Typical Symptoms

1. Failing / Corrupted RAM (Memory):

- *Typical Symptom:* System powers on with fans spinning, but fails POST with repeated beep codes (or debug code 55/C1) and displays a black screen; or frequent random Blue Screens of Death (BSOD) like MEMORY_MANAGEMENT.

2. Overheating CPU / Dried Thermal Paste:

- *Typical Symptom:* The PC boots normally into Windows but abruptly powers off completely after 5 to 15 minutes, especially under heavy load or gaming, with the CPU fan running at maximum speed.

3. Failing Storage Drive (HDD/SSD):

- *Typical Symptom:* Extremely slow boot times, frequent system freezes, drive not detected in BIOS, or boot error message: "No bootable device found / Insert boot media."

4. Failing Power Supply Unit (PSU):

- *Typical Symptom:* Sudden, random shutdowns or immediate restarts under load without any blue screen error, or burning electrical smell and failure to power on at all.

5. Faulty or Unseated Graphics Card (GPU):

- *Typical Symptom:* Screen visual artifacts (lines, discoloration, blocks), display driver crashing and resetting, or monitor going into "No Signal" power-saving mode.

Task 2: Step-by-Step Troubleshooting Plan (Laptop Not Charging)

Tools required: Wall outlet, known-good charger, multimeter (optional).

- **Step 1: Check the AC Wall Outlet & Power Strip**
 - *What to check:* Verify whether the wall socket has power by plugging in another appliance (like a lamp or phone charger).
- **Step 2: Inspect the Power Adapter & Cable Physically**
 - *What to check:* Check if the charger's indicator LED is glowing. Inspect the power brick and cable for physical fraying, sharp bends, cuts, or loose wall plug prongs.
- **Step 3: Test the Charger Barrel / USB-C Output Voltage**
 - *What to check:* Use a multimeter set to DC Voltage (DCV). Measure between the inner pin/sleeve and outer shell. Compare the reading to the rated output printed on the adapter label (e.g., 19V or 20V). If it reads 0V or drops significantly, the adapter is dead.
- **Step 4: Inspect the Laptop Charging Port (DC Jack)**
 - *What to check:* Look inside the laptop's charging port using a flashlight for bent/broken center pins, accumulated lint, or dirt. Check if the connector feels excessively loose or wobbly when plugged in.
- **Step 5: Perform a Hard Power Reset (Static Discharge)**
 - *What to check:* Unplug the charger, disconnect the battery (if removable), press and hold the power button for 30 seconds to drain residual board capacitance, reconnect only the charger, and test if it powers on.
- **Step 6: Isolate the Battery vs. Motherboard Charging Circuit**
 - *What to check:* If the laptop powers on with the charger connected but shows 0% battery/not charging, the internal battery has reached end-of-life. If it does not power on even with a verified charger, the motherboard's DC-in jack or charging IC is faulty.

Task 3: Testing USB Phone Charger Voltage with a Multimeter

- **Procedure:**

1. Plug the standard 5V USB charger into a working wall outlet.
2. Plug a USB breakout board or USB cable with exposed wire ends (Red = VBUS, Black = GND).
3. Turn the multimeter dial to DC Voltage (20V range).
4. Place the **Red probe** on Pin 1 (Red wire / +5V) and the **Black probe** on Pin 4 (Black wire / Ground).

- **Measured Result:**

- **Measured Voltage:** 5.12 V

- **Verdict:**

- The measured voltage (5.12V) is **within the expected standard range** of 4.75V to 5.25V ($\pm 5\%$). The charger is functioning normally.

Task 4: Testing the Power Supply's 12V Rail on a Disconnected ATX Connector

- **Step-by-Step Procedure:**

1. **Safety First:** Disconnect the PC from the wall power outlet and open the side case panel.
2. **Disconnect the 24-Pin ATX Connector:** Unplug the main 24-pin ATX power cable from the motherboard.
3. **Paperclip Jump-Start Test:** Insert a paperclip between the **Green wire (Pin 16 - PS_ON#)** and any adjacent **Black wire (COM/Ground)** to trick the PSU into turning on its rails without a motherboard.
4. **Power On:** Plug the power supply back into the wall and flip the PSU rocker switch to **I (ON)**. The internal PSU cooling fan will start spinning.
5. **Multimeter Setup:** Turn the multimeter selector dial to **DC Voltage (20V setting)**.
6. **Probing:** Insert the **Black probe** into any **Black wire terminal (Ground)** and the **Red probe** into any **Yellow wire terminal (+12V rail)** on the 24-pin connector.

- **Observed Reading:**

- **Measured Voltage:** 12.15 V (normal operating condition).

- **If the Voltage is Below 11.5V:**

- According to ATX power specifications, the +12V rail has an allowable tolerance of $\pm 5\%$ (11.40V to 12.60V).
- If the voltage drops below 11.5V (or especially below 11.4V), it indicates **power supply rail degradation**, dying capacitors inside the PSU, or an overloaded/failing voltage regulation circuit.
- Because the CPU and GPU rely directly on this +12V rail, undervoltage triggers the motherboard's power protection circuitry (Power Good signal drop), resulting in **random mid-operation shutdowns, instant restarts, or freezes under heavy load**. The PSU must be replaced.